



Cheryl A. Swanier · Voorhees University — CMP 224 — FacultyHack@Gateways 2026

EMPOWERING THE NEXT GENERATION OF PROGRAMMERS: AI AND JETSTREAM2 IN INTRODUCTORY COMPUTING



SGX3 is funded by the
National Science
Foundation
under award number
2231406.

ABSTRACT

The rapid advancement of artificial intelligence (AI) and cloud computing is transforming the way computer science is taught and practiced, creating new opportunities to provide students with authentic computational experiences early in their academic careers. This project explores the integration of AI-enabled learning tools and Jetstream2 cloud computing resources into an introductory programming course to enhance students' programming skills, computational thinking, and understanding of emerging technologies. The instructional model combines traditional programming concepts with AI-assisted code generation, debugging, code explanation, data analysis, and problem-solving activities within a cloud-based computing environment.

Jetstream2 provides students with access to scalable computational resources without requiring extensive local hardware or software configurations, thereby reducing technical barriers that can impede participation in introductory computing courses. Students use cloud-based virtual machines and development environments to write, execute, test, and evaluate programming solutions while using AI as a learning partner rather than a replacement for computational thinking. Instruction emphasizes responsible AI use, including evaluating AI-generated code for accuracy, identifying errors and hallucinations, addressing bias and ethical considerations, protecting data, and maintaining academic integrity.

The project also explores the use of Science Gateway resources to connect students to authentic computational tools, datasets, and research-oriented environments. Through hands-on activities and project-based learning, students progress from foundational programming concepts to cloud-based computational projects incorporating AI and data analysis. Student outcomes will be examined through programming assessments, project performance, surveys of confidence and engagement, and reflections on AI-supported learning. The anticipated outcome is an accessible and scalable instructional model that broadens participation in computing while preparing students with the technical and AI literacy skills needed for an increasingly cloud- and AI-driven workforce. This work demonstrates how Science Gateway resources can be leveraged to introduce beginning programmers to advanced computing environments while supporting student success and persistence in computer science.

INTRODUCTION

The target course is an introductory programming course designed for beginning computer science students, including students with limited or no prior programming experience. The course serves as a foundational course in the computing curriculum, introducing computational thinking, problem-solving, programming concepts, and emerging technologies that students will use in subsequent computer science courses. The project is motivated by the need to provide students with authentic exposure to high-performance/cloud computing, artificial intelligence, and modern development environments early in their academic careers. Integrating Jetstream2 and Science Gateway resources with AI-supported programming gives students access to scalable computing resources while reducing the technical barriers associated with configuring local environments. The project addresses the gap between traditional introductory programming instruction, and the increasingly AI- and cloud-driven computing environments students will encounter in advanced coursework and the workforce.

Course Goals

- Core Programming Concept:** Students will master fundamental programming concepts, including variables, data types, conditional statements, loops, functions, data structures, and basic object-oriented programming, using Python.
- Problem-Solving and Computational Thinking:** Students will develop computational thinking skills by decomposing real-world problems, designing algorithms, analyzing solutions, and using debugging and testing strategies to develop effective programs.
- Professional Best Practices:** Students will apply professional programming practices, including writing readable and well-documented code, using version control and collaborative tools, testing and debugging programs, documenting project processes, and developing reproducible computational workflows.
- Science Gateway / Jetstream2 / AI Integration:** Students will use **Jetstream2 cloud computing resources and Science Gateway tools**, including virtual machines and browser-based computational environments, to develop, execute, and evaluate JAVA programs and AI-supported computational projects while critically evaluating AI-generated code for accuracy, security, bias, and responsible use.

Grade Computation

Software Installation Readiness	10%
Lab Work	20%
Quizzes	10%
Programming Projects	30%
Exams	30%



Mentor Feedback Incorporated

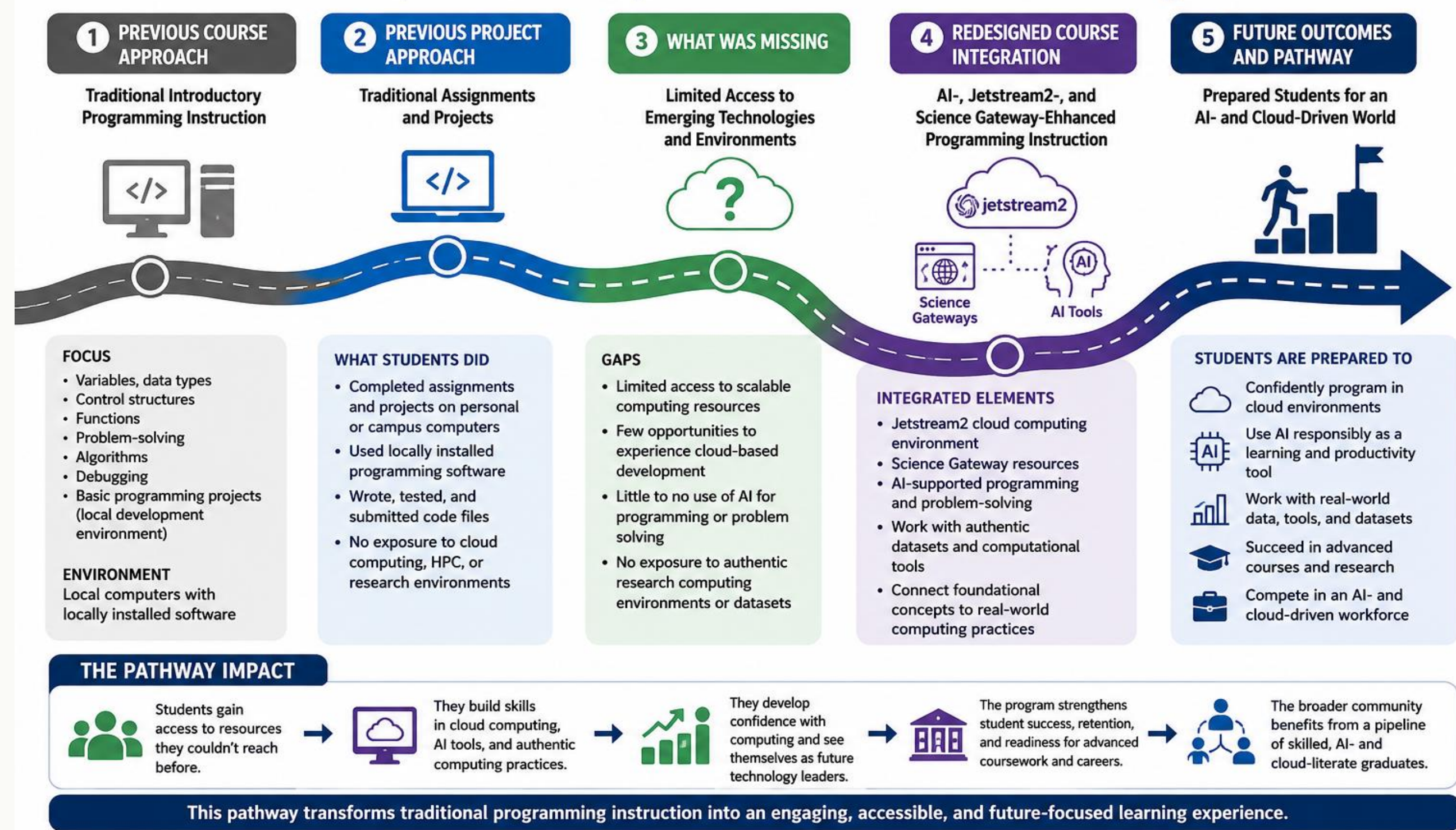
- Suggested that I used images for course redesign
- Suggested that I rewrite my project goals
- Suggested that I change the image for the Science Gateway Resources Used

IMPACT ON STUDENTS

- Primary users:** Beginning computer science students enrolled in the introductory programming course, particularly students with limited prior programming experience or limited access to advanced computing resources.
- Skills gained:** Students developed foundational JAVA programming, computational thinking, debugging, data analysis, cloud computing, and AI literacy skills.
- Increased access:** Students gained access to **Jetstream2 cloud computing resources** and **Science Gateway** environments that allowed them to work with computing resources beyond the capabilities of their personal computers.
- AI confidence:** Students learned to use **generative AI** as a programming and problem-solving partner while developing the ability to evaluate AI-generated code for accuracy, bias, security, and ethical concerns.
- Real-world experience:** Students moved beyond traditional local programming assignments and completed projects in a cloud-based environment similar to those used in research, industry, and advanced computing.

COURSE REDESIGN: BEFORE & AFTER

Pathway: From Traditional Programming Instruction to AI-, Jetstream2-, and Cloud-Enhanced Learning



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TARGET COURSE DESCRIPTION

Course Name	INTRO TO PROGRAMMING
Course Number	CMP 224
Department	COMPUTER SCIENCE
Enrollment	12 (Fall & Spring)
Prerequisites	NONE
Project Repo	https://github.com/cswanier/FacultyHack

Course Description

CMP 224 introduces problem solving and computer programming using the programming language Java. The course teaches how real-world problems can be solved computationally using the object-oriented metaphor that underlies Java. Some of the concepts and techniques covered in the course include data types, expressions, objects, methods, conditionals, iteration, program testing and debugging, interactive programs, data abstraction, interfaces, inheritance, polymorphism, and arrays. No previous programming experience is required; however, this course is intended for Computer Science majors or those who plan on applying to the major. Non-majors are strongly encouraged to take one of our programming courses designed for non-majors. Use of a laptop computer on which you can install software is required.

SCIENCE GATEWAY RESOURCES USED

Science Gateway resources will be used to connect students to accessible, research-oriented computational tools and environments that complement Jetstream2. Jetstream2 itself supports science gateways and provides web-based interfaces and preconfigured virtual machines that simplify access to advanced computing resources.

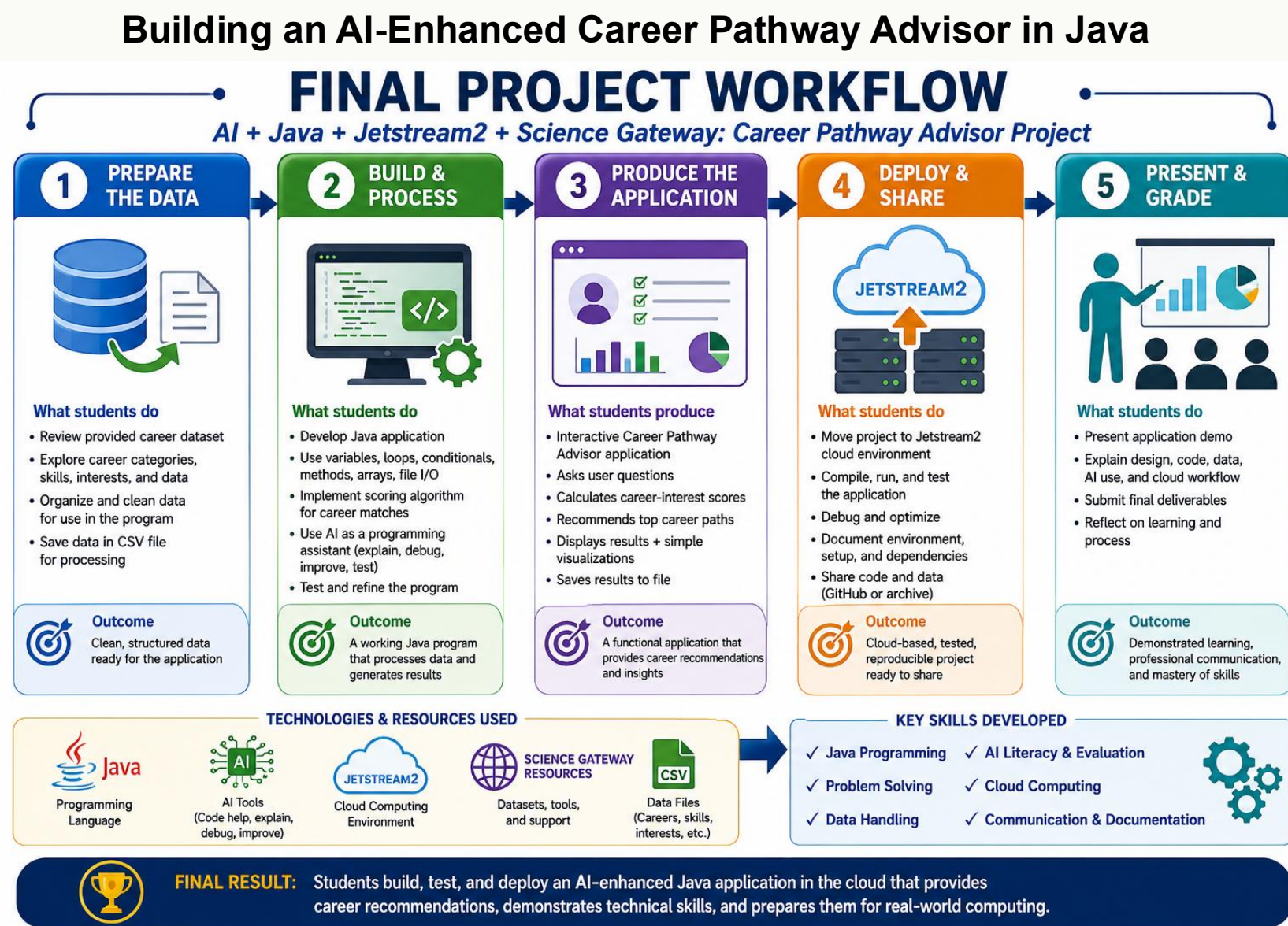
Jetstream2 is a cloud-based computing environment funded by the National Science Foundation (NSF) and designed to support research and education across a wide range of scientific disciplines. It is a part of the **XSEDE** (Extreme Science and Engineering Discovery Environment) and is operated by Indiana University in collaboration with several partner institutions.

In this course, **Jetstream2** is being used to **provide students with consistent, on-demand computing environments** for coding, testing, debugging, and running Java programs and other software tools. This approach eliminates issues related to hardware or software differences on personal computers and ensures that every student can access the same setup, regardless of their device.

By working with Jetstream2, students will **also gain experience using cloud infrastructure**—an increasingly important skill in both academia and industry. Through this platform, students will learn how to **interact with virtual machines, manage computing environments, and explore how scalable systems can support software development and data processing tasks**.



FINAL PROJECT / DELIVERABLE



Project Goals

- Goal 1:** Increase access to computing resources with Jetstream2 virtual computing environments
 - Goal 2:** Integrate generative AI into programming instruction for code generation, debugging, explanation, and problem solving.
 - Goal 3:** Develop authentic computational projects using Jetstream2 and AI that mirror real-world computing applications.
- Science Gateway Goal:** Integrate Science Gateway resources, including Jetstream2 cloud computing infrastructure and AI-enabled tools, into an introductory programming course to provide students with accessible, authentic, and scalable computing experiences that strengthen computational thinking, programming skills, and AI literacy.

EXPANSION IDEAS

- Develop undergraduate research opportunities** that allow students to use Jetstream2 and AI for authentic research and computational projects.
- Offer faculty development workshops** to help instructors learn how to incorporate AI, cloud computing, and Science Gateway resources into their courses.
- Create summer and bridge programs** that introduce high school and incoming college students to cloud computing, programming, and AI.
- Establish institutional partnerships** with other HBCUs, minority-serving institutions, and research universities to share resources, curriculum, and best practices.

REFERENCES

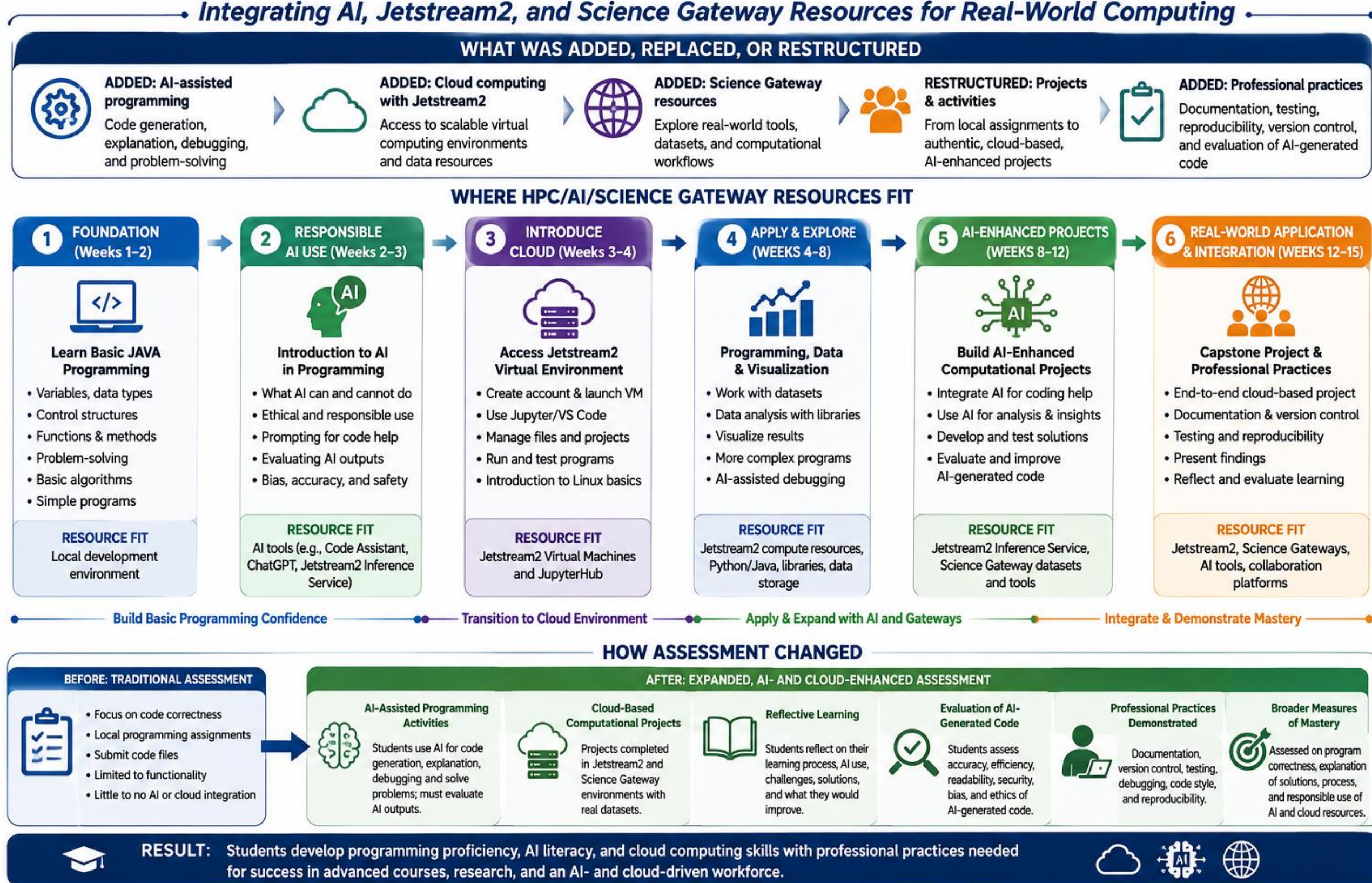
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LESSONS LEARNED

- What worked:** A major strength of the project was the progressive introduction of technology. Rather than asking students to begin immediately with cloud computing or AI, the course first established foundational Java programming skills and then introduced responsible AI use before transitioning students to Jetstream2.
- What I would change:** One anticipated friction point is the **initial technical setup and access to Jetstream2**, particularly account creation, authentication, virtual machine configuration, and learning how to navigate a cloud environment.
- What surprised me:** What was most surprising was the potential for **AI to change students' attitudes toward programming**.

Key Content by Week

PATHWAY: REDESIGNING INTRODUCTORY PROGRAMMING INSTRUCTION



PROJECT GITHUB REPO

[github.com/github-user/your-repo]

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